Exact Survival Laws and Orbit Bounds for Line Units in Binary Group Algebras

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Abstract

Let $q = 2^n$ and consider units of augmentation one supported on lines in $\mathbb{F}_q^\rho$, inside the binary group algebra $\mathbb{F}_2[\mathbb{F}_q^\rho]$. We study two questions: the probability that a random product of line units fixes a coefficient, and the probability that the associated multilinear Boolean evaluation is constant along a target line. Three distinct coplanar lines admit a fixed coefficient of minimum possible support four. Passing to an additive quotient turns the fixing equation into a product equation among normalized units, which gives exact survival probabilities for one or two uniform units on arbitrary lines of the plane and for any number of units on the three defining lines; in particular survival probabilities of independent units need not multiply. For a singleton coefficient and a fixed transverse line-sum unit, an exposed-column rank argument gives an orbit-constancy bound $\exp(-\Omega(\sqrt{\mathcal{A}}))$ with $\mathcal{A} = q2^{-(\rho-1)}/s$, uniform over coordinate-full directions; it vanishes at every fixed density $\rho/n \to \varkappa < 1$. For the repeated-slope family only (not for general directions), the orbit-constancy probability has the exact critical limit $e^{-\lambda}$ as $n, \rho \to \infty$ with $q2^{-(\rho-1)} \to \lambda \in [0, \infty)$. Positive association gives lower bounds for four-point coefficients that stay positive at $\rho = 2n + O(1)$ for arbitrary target lines and at $\rho = n + O(1)$ for units on the target line. All proofs are complete and self-contained; small cases are checked exhaustively by supplementary scripts, including an exact count over all 224 normalized slopes over $\mathbb{F}_{16}$. Limitations: the generic critical window $\varkappa = 1$, the true cross-line transition, and survival laws for three general lines remain open. The work was produced with AI assistance and has not been reviewed by experts.

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Problem and contribution

Let $q = 2^n$ and $G = \mathbb{F}_q^\rho$, viewed as an additive group, and let $\mathbb{F}_2[G]$ be its binary group algebra. An augmentation-one element supported on a line (a one-dimensional $\mathbb{F}_q$-subspace) is a unit, and every such unit is an involution. This paper studies two elementary but distinct questions about multiplication by such line units: (i) when does a random product of line units fix a given coefficient, and (ii) when is the associated multilinear Boolean evaluation constant along the translates of a target line? The point of the paper is that these two questions have exact or sharp answers in several natural regimes, and that the answers differ.

Unconditional results (all proved in full below).

- A support-four coefficient $K = \Omega_H$, built from three distinct coplanar lines, is fixed by the product of their line-sum units, and every nonzero coefficient fixed by that product has support at least four (Proposition 3.1).
- Exact survival probabilities: for one uniform line unit, for two independent uniform line units on arbitrary lines in the plane (Theorem 4.2), and for any number of independent uniform units on the three defining lines, including repetitions (Theorem 4.3). For arbitrary families of supporting lines the joint survival probability is $1/|P|$ for an explicit subgroup P of the quotient unit group, hence at least the product of the individual probabilities and unchanged by repeated lines (Proposition 4.4); the two- and three-line laws show that the inequality can be strict.
- For a singleton coefficient and a fixed transverse line-sum unit, a uniform bound $\mathbb{P}[\text{orbit constant}] \le \exp(-\Omega(\sqrt{\mathcal{A}}))$ with $\mathcal{A} = q2^{-(\rho-1)}/s$, valid for all coordinate-full directions, hence vanishing at every fixed density $\rho/n \to \varkappa < 1$ (Theorem 5.2).

- For a repeated-slope family, the orbit probability equals the empty-active-set probability $(1 - 2^{-(\rho-1)})^q$ up to an additive error $1/(2q)+e^{-q/72}+2^{1-q/6}$, uniformly in the parameters, giving the critical limit $e^{-\lambda}$ when $q2^{-(\rho-1)} \to \lambda$ (Theorem 6.1).
- Lower bounds from positive association for four-point coefficients: nonvanishing at $\rho = 2n + O(1)$ for arbitrary target lines, with the sharper limit $e^{-2^{-c}}$ for every coefficient on a second line of the defining plane and at $\rho = n + O(1)$ when the unit lies on the target line (Theorem 7.2, Corollary 7.3).

Not claimed. We do not determine the orbit transition for generic directions at critical density $\kappa = 1$, nor the true cross-line transition for four-point coefficients (the scale $\rho = 2n$ is only where our lower bound becomes nonvanishing). We do not claim exact survival laws for three general supporting lines. The finite computation in Section 8 checks one finite instance of the counting step; it is not part of any proof.

Related work and distinction

Normalized units of modular group algebras of p-groups are classical objects; in characteristic two for an elementary abelian 2-group, the identity $B^2 = \mathrm{aug}(B)1$ (Lemma 2.1) is standard, and we include its one-line proof only to fix notation. Borello, Willems and Zini [1] relate the dimension and minimum support of ideals in group algebras (an uncertainty principle $d(C)\dim C \geq |G|$) and study Schur (coordinatewise) products of ideals. The ideal $\Omega_H\mathbb{F}_2[G]$ generated by our four-point coefficient has minimum distance 4 and dimension $|G|/4$, so it is an equality case of their bound [1] (Corollary 2.6), of the form classified in their Theorem 2.10; Proposition 3.1 adds that the strictly larger annihilator $\mathrm{Ann}(D-1)$, of dimension at least $|G|/2$, has the same minimum distance. Our survival laws concern convolution products of random units and an additive quotient that turns the fixed-coefficient equation into a product equation among subalgebras. The orbit lower bounds use the Harris–FKG positive association inequality for product measures [2], [3], for which a short proof of the needed special case is included. The orbit upper bound is a rank bound for a structured random matrix over $\mathbb{F}_2$; Darling, Penrose, Wade and Zabell [4] (Section 2.1) study sparse random $\mathbb{F}_2$ matrices with i.i.d. rows, each sampled uniformly after its weight is specified; the rows of our matrix instead share role variables through field-index collisions, so row independence is unavailable, and Lemma 5.1 extracts columns supported on disjoint sets of role variables. The full-support line model is the one appearing in Reed–Solomon evaluation vectors [5]; this is a motivation only.

1. Introduction

Convolution by a line-supported element is a simple operation in a finite group algebra. Its effect on coefficient supports is algebraic, while its effect on a multilinear Boolean evaluation depends on the number and arrangement of the underlying role variables. We determine several exact survival laws and orbit bounds that make this distinction quantitative. The central finite object is a four-point additive subgroup meeting three coplanar lines in two points each. Passing to its quotient turns the survival equation into a product equation among normalized units.

The algebraic results apply to arbitrary lines. The orbit upper bound requires directions with no zero coordinates, as in the full-support line model associated with Reed–Solomon evaluations [5]. To see the connection explicitly, for distinct evaluation points $\xi_1, \dots, \xi_\rho$ and a polynomial f with $f(\xi_j) \neq 0$, scalar multiplication of the evaluation vector gives the line $\{g(f(\xi_1), \dots, f(\xi_\rho)) : g \in \mathbb{F}_q\}$. We prove the orbit bound for every nonzero-coordinate direction, without assuming a polynomial degree restriction. No such interpolation interpretation is needed for the group-algebra statements.

The main results have three forms. Theorems 4.2 and 4.3 count exactly the random units preserving the four-point coefficient. Theorem 5.2 bounds the probability of a constant orbit for a singleton and a fixed transverse line-sum unit, uniformly in the direction. Theorem 7.2 gives lower bounds for the orbit probability of a four-point coefficient multiplied by a random line unit. A structured singleton family also admits the exact critical limit in Theorem 6.1.

Normalized units and ideals in modular group algebras provide the natural algebraic setting. Related work of Borello, Willems and Zini [1] connects ideal dimension with minimum support and studies Schur products of ideals; the remark after Proposition 3.1 places our four-point coefficient precisely relative to their uncertainty principle. The survival equations here use convolution and intersections of subalgebras after an additive quotient. The probability lower bound uses the classical positive association of product measures [2], [3]; a proof of the precise form needed here is included. The independent-column argument is also elementary, but its applicability depends on keeping the role labels and affine-index collisions explicit. Rank deficiency of sparse random matrices over $\mathbb{F}_2$ is studied, for example, by Darling,

Penrose, Wade and Zabell [4] (Section 2.1), for matrices with i.i.d. rows, each row sampled uniformly after its weight is specified. The rows of the matrices H_X below instead share role variables through affine-index collisions, so row independence is unavailable; Lemma 5.1 handles the dependence by extracting columns supported on disjoint sets of role variables.

2. Group algebras, fixed coefficients and orbit events

Throughout, $q = 2^n$ and $G = \mathbb{F}_q^\rho$ is regarded as an additive group. The binary group algebra $\mathbb{F}_2[G]$ has basis $[g]$, $g \in G$, with $[g][h] = [g + h]$ and identity $1 = [0]$. For $B = \sum_g b_g[g]$, write $\mathrm{aug}(B) = \sum_g b_g \in \mathbb{F}_2$ and $\mathrm{supp}(B) = \{g : b_g = 1\}$. For an additive subgroup $S \le G$, put

$$\Omega_S = \sum_{s \in S} [s], \qquad \mathcal{U}(S) = \{B \in \mathbb{F}_2[S] : \mathrm{aug}(B) = 1\}.$$

All algebraic sums and products of coefficients in this paper are over $\mathbb{F}_2$; probabilities and cardinalities are ordinary real numbers and integers.

Lemma 2.1. For every $B \in \mathbb{F}_2[G]$, $B^2 = \mathrm{aug}(B)1$. Consequently $\mathcal{U}(S)$ is a group of order $2^{|S|-1}$, and every element of $\mathcal{U}(S)$ is its own inverse. For additive subgroups $S, T \le G$,

$$\Omega_S \Omega_T = |S \cap T|\, \Omega_{S+T}, \tag{1}$$

where the integer coefficient on the right is reduced modulo two.

Proof. Commutativity and characteristic two cancel all cross terms in B^2; each remaining basis square is $[2g] = 1$. Augmentation is a nonzero linear functional on the $|S|$-dimensional coefficient space, which gives the cardinality. Every element of $S + T$ has exactly $|S \cap T|$ representations as $s + t$, proving (1). □

A line is a one-dimensional $\mathbb{F}_q$-subspace. We call its direction *coordinate-full* if all coordinates of a nonzero generator are nonzero. For a line L, its line-sum unit is $1 + \Omega_L$; it has augmentation one and support $L \setminus \{0\}$.

Definition 2.2. A nonzero coefficient $K \in \mathbb{F}_2[G]$ is *fixed* by a unit D if $KD = K$. Its survival under a further unit A is the event $KDA = K$.

Let $(X_{j,u} : 1 \le j \le \rho,\ u \in \mathbb{F}_q)$ be independent uniform bits. The Boolean phase of B is

$$Q_B(X) = \sum_{g \in G} b_g \prod_{j=1}^{\rho} X_{j,g_j} \in \mathbb{F}_2. \tag{2}$$

Translation is $\tau_t B = [t]B$. For a target line T, let

$$\mathcal{C}_T(B) = \{X : Q_{\tau_t B}(X) \text{ is constant as } t \text{ ranges over } T\}.$$

This event allows both constant values. It is not the algebraic equation $BA = B$, and no identification between these two conditions will be used.

3. A support-minimal four-point fixed coefficient

Let $q \ge 4$ and let $W \le G$ be a two-dimensional $\mathbb{F}_q$-subspace. Take distinct lines $L_1, L_2, L_3 \le W$. There are linearly independent $a, b \in W$ with $a \in L_1$, $b \in L_2$ and $a + b \in L_3$. Define

$$H = \{0, a, b, a + b\}, \qquad K = \Omega_H = (1 + [a])(1 + [b]), \qquad D = \prod_{i=1}^{3} (1 + \Omega_{L_i}). \tag{3}$$

Proposition 3.1. The coefficient K satisfies $KD = K$. Every nonzero coefficient fixed by D has support at least four, even if its support is allowed outside W.

Proof. Each $H \cap L_i$ has size two, so (1) gives $K\Omega_{L_i} = 0$. Thus K is fixed by each factor of D. For distinct lines in W, their subgroup sums multiply to Ω_W; a product of three such sums vanishes because $|W \cap L_i| = q$ is even. Consequently

$$N := D - 1 = \Omega_{L_1} + \Omega_{L_2} + \Omega_{L_3} + \Omega_W.$$

Its support is $W \setminus (L_1 \cup L_2 \cup L_3)$, of size $(q-1)(q-2) > 0$. A coefficient of odd support is a unit by Lemma 2.1, and hence cannot annihilate N.

A two-point coefficient, after translation, is $1 + [h]$ with $h \neq 0$. If $h \notin W$, then N and $[h]N$ have disjoint supports. If $h \in W$ and $[h]N = N$, the union $S = L_1 \cup L_2 \cup L_3$ is invariant under translation by h. Since $0 \in S$, this forces $h \in S$, say $h \in L_1$. Choose a selected line $L_2 \neq L_1$ and an unselected line $M \leq W$. The affine line $h + L_2$ intersects M at a nonzero point: these lines are not parallel, and $0 \notin h + L_2$. That point is outside S, contradicting $h + L_2 \subseteq S$. This excludes support two and proves the asserted minimum. □

The proof uses no restriction on coordinates. In particular it applies whenever the three selected lines are coordinate-full.

The principal ideal $J_H = \Omega_H \mathbb{F}_2[G]$ has a basis of the disjoint coset sums Ω_{g+H}, so $\dim_{\mathbb{F}_2} J_H = |G|/4$ and every nonzero element has support a positive multiple of four. Hence J_H attains equality in the bound $d(C) \dim C \geq |G|$ for ideals of group algebras [1] (Corollary 2.6), and is of the form classified in [1] (Theorem 2.10). Proposition 3.1 concerns the annihilator $C_D = \mathrm{Ann}(D-1)$, which contains J_H because $KN = 0$. Since $D^2 = 1$ we have $N^2 = 0$, so the image of multiplication by N lies in its kernel and $\dim C_D \geq |G|/2 > \dim J_H$. The proposition shows that this strictly larger ideal still has minimum distance four. In the rest of the paper, the notation H, K, D, W refers to (3).

4. Exact survival laws from an additive quotient

Let $\pi : W \to Q = W/H$ be the quotient of additive groups; it is generally not a quotient of $\mathbb{F}_q$-vector spaces. Its cardinality is $|Q| = q^2/4$. Extend π linearly to $\pi_* : \mathbb{F}_2[W] \to \mathbb{F}_2[Q]$.

Lemma 4.1. For every $B \in \mathbb{F}_2[W]$,

$$KB = K \quad \Longleftrightarrow \quad \pi_*(B) = 1. \tag{4}$$

If $M \leq W$ is a line, $h = |H \cap M|$ and $S = \pi(M)$, then $h \in \{1, 2\}$, $|S| = q/h$, and every fiber of $\mathbb{F}_2[M] \to \mathbb{F}_2[S]$ has cardinality

$$2^{q-q/h}. \tag{5}$$

The pushforward of the uniform measure on $\mathcal{U}(M)$ is uniform on $\mathcal{U}(S)$.

Proof. The translates $K[g] = \Omega_{H+g}$ have disjoint supports for distinct H-cosets. Thus KB records the parity of the coefficients of B in each such coset, proving (4). The nonzero elements of H lie on three distinct $\mathbb{F}_q$-lines, so h is one or two. The coefficient pushforward is surjective and has rank $|S| = q/h$, hence kernel dimension $q - q/h$. It preserves augmentation. Therefore the entire fiber above an augmentation-one target lies in the augmentation-one source, and all these fibers have the size in (5). Notice that this coefficient-space fiber is larger than the point-map fiber of size h when $h = 2$ and $q > 2$. □

Since $KD = K$, survival under DA is precisely $KA = K$. For a uniform $A \in \mathcal{U}(M)$, Lemma 4.1 immediately gives

$$\mathbb{P}[KA = K] = 2^{1-q/|H \cap M|}. \tag{6}$$

Theorem 4.2 (Two line units). Let A_i be independent uniform elements of $\mathcal{U}(M_i)$, where $M_1, M_2 \leq W$ are lines, and write $h_i = |H \cap M_i|$. If $M_1 \neq M_2$,

$$\mathbb{P}[KA_1A_2 = K] = 2^{1+4/(h_1h_2)-q/h_1-q/h_2}. \tag{7}$$

If $M_1 = M_2 = M$, this probability is $2^{1-q/|H \cap M|}$.

Proof. Set $S_i = \pi(M_i)$ and $x_i = \pi_*(A_i)$. These are independent uniform elements of $\mathcal{U}(S_i)$. For distinct lines, $S_1 + S_2 = Q$ and

$$|S_1 \cap S_2| = \frac{|S_1||S_2|}{|Q|} = \frac{4}{h_1h_2}.$$

By Lemma 2.1, $x_1x_2 = 1$ if and only if $x_1 = x_2$. Their common values are exactly $\mathcal{U}(S_1 \cap S_2)$, because the intersection of two group-basis subalgebras is the subalgebra on the intersection of their supports. There are $2^{|S_1 \cap S_2|-1}$ such values, out of $2^{|S_1|+|S_2|-2}$ ordered pairs. This proves (7). For a repeated line there is one admissible second unit for each first unit. □

The excess over the product of the two one-unit probabilities is $2^{|S_1\cap S_2|-1}$. It equals 8, 2 or 1 when (h_1, h_2) is respectively $(1,1)$, $(1,2)$ or $(2,2)$, with the mixed case symmetric. For the three defining lines, Theorem 4.3 gives the analogous excess d over the product $2^{3(1-d)}$ of the three one-unit probabilities. Independent sampling of units therefore need not give multiplicative survival probabilities.

Theorem 4.3 (Units on the three defining lines). Let $A_1, \dots, A_k$ be independent uniform line units, with $k \geq 1$ and each supporting line among L_1, L_2, L_3. Let j be the number of distinct supporting lines and put $d = q/2$. Then

$$\mathbb{P}\left[K\prod_{i=1}^{k} A_i = K\right] = \begin{cases} 2^{1-d}, & j = 1, \\ 2^{2-2d}, & j = 2, \\ d\,2^{-3(d-1)}, & j = 3. \end{cases} \tag{8}$$

The probability depends on the active lines and not on their multiplicities.

Proof. Using a, b as an $\mathbb{F}_q$-basis identifies Q with $(\mathbb{F}_q/\mathbb{F}_2)^2$. The three images S_1, S_2, S_3 are the two coordinate axes and the diagonal, each of size d. Suppose $x_i \in \mathcal{U}(S_i)$ and $x_1x_2x_3 = 1$. Then $x_1x_2 = x_3$, and the support of x_1x_2 is the Cartesian product of the nonempty supports of x_1 and x_2. There is no cancellation: every element of $S_1 \oplus S_2$ has a unique decomposition into the two axes. A nonempty Cartesian product contained in the diagonal is a single point. The admissible triples are consequently

$$([(u,0)],[(0,u)],[(u,u)]), \qquad u \in \mathbb{F}_q/\mathbb{F}_2.$$

Their number is d, proving the $j = 3$ case. The first two cases follow from (6) and Theorem 4.2. Finally, the product of $k' \geq 1$ independent uniform elements of any finite group is uniform: each target has exactly $|\mathcal{U}(S_i)|^{k'-1}$ preimages. Group the factors on each supporting line to obtain the general statement. □

The excesses above have a common source, valid for arbitrary supporting lines.

Proposition 4.4 (Arbitrary families of supporting lines). Let $M_1, \dots, M_k \leq W$ be lines, not necessarily distinct, and let A_i be independent uniform elements of $\mathcal{U}(M_i)$. Put $V_i = \mathcal{U}(\pi(M_i))$ and let $P = V_1 \cdots V_k$ be the subgroup of $\mathcal{U}(Q)$ they generate. Then

$$\mathbb{P}\Big[K\prod_{i=1}^{k} A_i = K\Big] = \frac{1}{|P|} \geq \prod_{i=1}^{k} \mathbb{P}[KA_i = K],$$

and the ratio of the two sides is $|\ker\Phi|$, where $\Phi : V_1 \times \cdots \times V_k \to P$ is multiplication. In particular, the probability is a power of 2, it is unchanged when a supporting line is repeated, and it does not increase when an independent uniform unit on a further line is added.

Proof. By Lemma 4.1 the elements $x_i = \pi_*(A_i)$ are independent and uniform on V_i, and survival is the event $x_1 \cdots x_k = 1$. The group $\mathcal{U}(Q)$ is abelian, so Φ is a surjective homomorphism onto P and all its fibers have size $|\ker\Phi|$. Hence $x_1 \cdots x_k$ is uniform on P, which gives the equality. Since $\mathbb{P}[KA_i = K] = 1/|V_i|$ and $|P|\,|\ker\Phi| = \prod_i |V_i|$, the inequality and the ratio follow. The group P is a 2-group, it does not change when a factor V_i is repeated, and it can only grow when a factor is added. □

For two distinct lines $|\ker\Phi| = |V_1 \cap V_2| = 2^{|\pi(M_1)\cap\pi(M_2)|-1}$, recovering Theorem 4.2; for the three defining lines the ratio is the factor d in (8). The proposition does not compute $|P|$ for three general supporting lines.

5. Exposed columns for a singleton line-sum orbit

In this section the coefficient is the singleton $[0]$, and the unit is the fixed line sum $U = 1 + \Omega_L$, not a uniformly sampled line unit. Let $\rho \geq 2$ and take two distinct coordinate-full lines. Independent coordinate scalings preserve the distribution of the role arrays, so we may normalize them to

$$T = \{a\mathbf{1} : a \in \mathbb{F}_q\}, \qquad L = \{gv : g \in \mathbb{F}_q\}, \qquad v \in (\mathbb{F}_q^*)^\rho.$$

The nonparallel assumption is $v \notin \mathbb{F}_q\mathbf{1}$. Write $Y_x = X_{\rho,x}$ and $d_j = v_j/v_\rho$ for $j < \rho$. Let

$$J = \{j < \rho : d_j \neq 1\}, \quad s = |J| \geq 1, \qquad I = \{j < \rho : d_j = 1\}, \quad t = |I|, \quad p = 2^{-(\rho-1)},$$

and let $\nu = |\{d_j : j \in J\}|$ be the number of distinct ratios different from one, so that $1 \le \nu \le s$. Changing variables from g to $x = a + v_\rho g$ gives

$$z(a) := Q_{\tau_{a1}U}(X) = \sum_{x \in \mathbb{F}_q} H_X(a,x) Y_x, \qquad H_X(a,x) = \mathbf{1}_{x \neq a} \prod_{j<\rho} X_{j,(1+d_j)a+d_j x}. \tag{9}$$

The zero diagonal in this definition is essential.

Lemma 5.1 (Exact exposed-column probability). Fix $E \subseteq \mathbb{F}_q$ with $|E| = M \le q/2$ and a nonzero relation $c \in \mathbb{F}_2^E$ with support C of size m. There is an exposed set $F \subseteq \mathbb{F}_q \setminus E$, chosen independently of the role arrays, with

$$|F| \ge R_M := \left\lfloor \frac{q-M}{\nu M(M-1)+1} \right\rfloor, \tag{10}$$

and, if E is an $\mathbb{F}_2$-linear subspace of $\mathbb{F}_q$,

$$|F| \ge R_M^{\mathrm{sub}} := \left\lfloor \frac{q-M}{\nu(M-1)+1} \right\rfloor, \tag{11}$$

such that

$$\mathbb{P}_X[(H_X^{\mathsf{T}} c)|_F = 0] = \left(1 - 2^{-t}\,\frac{1-(1-2^{1-s})^m}{2}\right)^{|F|} \le (1-p)^{|F|}. \tag{12}$$

Proof. For $j \in J$, let $\ell_j(a,x) = (1+d_j)a + d_j x$. A collision $\ell_j(a,x) = \ell_j(b,y)$ in the same role, with $x \neq y$, requires

$$y = x + d_j^{-1}(1+d_j)(a+b), \qquad a \neq b. \tag{13}$$

Make a graph on $\mathbb{F}_q$ using every edge in (13) for $j \in J$ and distinct $a, b \in E$, and remove E. The edges in (13) depend on j only through d_j, so roles with equal ratios contribute the same edges, and the shifts are $d_j^{-1}(1+d_j)e$ with $e \in (E+E)\setminus\{0\}$. The maximum degree is therefore at most $\nu|(E+E)\setminus\{0\}| \le \nu M(M-1)$; when E is a subspace, $(E+E)\setminus\{0\} = E\setminus\{0\}$ and the bound is $\nu(M-1)$. Greedy selection of a vertex and deletion of its neighbors produces an independent set satisfying (10), and (11) in the subspace case. The parameters s and t in (12) remain the numbers of roles, not of distinct ratios.

For $x \in F$ and $a \in C$, the products

$$Q_{a,x} = \prod_{j \in J} X_{j,\ell_j(a,x)}$$

are independent Bernoulli variables of parameter 2^{-s}. Within one column, $a \mapsto \ell_j(a,x)$ is injective for $j \in J$; between columns, independence follows from the forbidden-edge construction. Variables in different roles remain distinct even when their field indices agree. The remaining roles contribute the common factor $R_x = \prod_{j \in I} X_{j,x}$, a Bernoulli variable of parameter 2^{-t}, independent across columns and from all the $Q_{a,x}$. The empty product is one. Since F avoids E,

$$(H_X^{\mathsf{T}} c)_x = R_x \sum_{a \in C} Q_{a,x}.$$

The parity of m independent Bernoulli(2^{-s}) variables is one with probability $[1-(1-2^{1-s})^m]/2 \ge 2^{-s}$. All exposed column values are independent, proving both parts of (12). □

Theorem 5.2 (Uniform averaged row-rank bound). With the preceding notation, for every fixed E of size $M \le q/2$,

$$\mathbb{P}_X[\mathrm{rank}_{\mathbb{F}_2} H_X[E,*] < M] \le (2^M - 1)(1-p)^{R_M} \le 2^M e^{-pR_M}, \tag{14}$$

$$\mathbb{P}[\mathcal{C}_T(U)] \le 2^M e^{-pR_M} + 2^{1-M}, \tag{15}$$

and both bounds hold with R_M^{sub} in place of R_M when E is an $\mathbb{F}_2$-subspace. Let $\mathcal{A} = qp/\nu$. If $\mathcal{A} \ge 64$, then

$$\mathbb{P}[\mathcal{C}_T(U)] \le e^{-(1-(\log 2)/4)\sqrt{\mathcal{A}}} + 2e^{-(\log 2)\sqrt{\mathcal{A}}/8} = \exp\bigl(-\Omega(\sqrt{\mathcal{A}})\bigr), \tag{16}$$

uniformly over the allowed directions. In particular, $2^{n-\rho}/\rho \to \infty$ is sufficient for the orbit probability to tend to zero, and this includes every fixed density $\rho/n \to \kappa < 1$.

Proof. Rank deficiency gives a nonzero row relation. Apply Lemma 5.1 to each of the $2^M - 1$ relations and take a union bound. Conditional on full row rank, the uniform vector Y maps uniformly onto $\mathbb{F}_2^M$, where the two constant vectors have total probability 2^{1-M}. This proves (15).

For (16), let E be an $\mathbb{F}_2$-subspace of cardinality $M = 2^{\lfloor \log_2(\sqrt{\mathcal{A}}/4) \rfloor}$, so that $\sqrt{\mathcal{A}}/8 < M \le \sqrt{\mathcal{A}}/4$; in particular $M \ge 2$ and, since $\mathcal{A} \le q$, $M \le q/2$. As $\nu(M-1)+1 \le \nu M$ and $q - M \ge q/2$,

$$R_M^{\mathrm{sub}} \ge \frac{q}{2\nu M} - 1 \ge \frac{q}{4\nu M},$$

where the last step uses $q/(4\nu M) \ge q/(\nu\sqrt{\mathcal{A}}) = \sqrt{\mathcal{A}}/p \ge 1$. Hence $pR_M^{\mathrm{sub}} \ge \mathcal{A}/(4M) \ge \sqrt{\mathcal{A}}$, and

$$2^M e^{-pR_M^{\mathrm{sub}}} \le e^{-(1-(\log 2)/4)\sqrt{\mathcal{A}}}, \qquad 2^{1-M} \le 2e^{-(\log 2)\sqrt{\mathcal{A}}/8}.$$

Finally $\nu \le s \le \rho - 1$, so the stated sufficient density condition forces $\mathcal{A} \to \infty$. □

6. An exact critical limit for a repeated-slope family

The uniform upper bound leaves a critical window. A family with repeated slopes provides a solvable example within that window. Fix $\gamma \in \mathbb{F}_q^* \setminus \{1\}$ and $\rho \ge 3$, and take

$$T = \mathbb{F}_q(1, \dots, 1), \qquad L = \mathbb{F}_q(1, \dots, 1, \gamma), \qquad U = 1 + \Omega_L.$$

Let $p = 2^{-(\rho-1)}$ and define the random active set $\Lambda = \{h : \prod_{j<\rho} X_{j,h} = 1\}$. Its membership indicators are independent Bernoulli(p) variables. Direct substitution gives

$$z(a) = \sum_{\substack{h \in \Lambda \\ h \ne a}} Y_{a+\gamma(h-a)}. \tag{17}$$

Theorem 6.1 (Repeated-slope Poisson limit). Uniformly over all $\rho \ge 3$ and all $\gamma \in \mathbb{F}_q^* \setminus \{1\}$,

$$0 \le \mathbb{P}[\mathcal{C}_T(U)] - (1-p)^q \le \frac{1}{2q} + e^{-q/72} + 2^{1-q/6}. \tag{18}$$

If $n \to \infty$, $\rho \to \infty$ and $qp \to \lambda \in [0, \infty)$, then

$$\mathbb{P}[\mathcal{C}_T(U)] \longrightarrow e^{-\lambda}. \tag{19}$$

If $qp \to \infty$, the probability tends to zero.

Proof. The lower bound is the event $\Lambda = \varnothing$. Let H_Λ be the matrix in (17). Adding the diagonal matrix of $\mathbf{1}_\Lambda$ gives the matrix $B_\Lambda(a, x) = \mathbf{1}_\Lambda(\gamma^{-1}x + (1+\gamma^{-1})a)$. Permuting rows and columns changes B_Λ into the additive convolution matrix for $t_\Lambda = \sum_{h\in\Lambda}[h]$. Thus

$$\operatorname{rank}_{\mathbb{F}_2} H_\Lambda \ge \operatorname{rank}_{\mathbb{F}_2}(t_\Lambda \cdot) - |\Lambda|. \tag{20}$$

Write $\epsilon = |\Lambda| \bmod 2$ and $\ell = \sum_{h\in\Lambda} h \in \mathbb{F}_2^n$. When $\epsilon = 1$, Lemma 2.1 gives rank q for multiplication by t_Λ. When $\epsilon = 0$ and $\ell \ne 0$, use

$$\mathbb{F}_2[\mathbb{F}_2^n] \cong \mathbb{F}_2[u_1, \dots, u_n]/(u_1^2, \dots, u_n^2), \qquad u_i = 1 + [e_i].$$

The degree-one part of t_Λ is nonzero. An invertible linear substitution makes it u_1; such a substitution is an algebra automorphism, because the square of every linear form in the u_i vanishes in characteristic two. Multiplication by t_Λ is then injective on the $q/2$-dimensional span of monomials not containing u_1: the lowest nonzero homogeneous part of any element of this span is carried to its nonzero product with u_1. Hence its rank is at least $q/2$.

On $|\Lambda| \le q/3$ and $(\epsilon, \ell) \ne (0,0)$, (20) gives $\operatorname{rank}_{\mathbb{F}_2} H_\Lambda \ge q/6$. As above, conditional orbit constancy has probability at most $2^{1-q/6}$. Since $p \le 1/4$, the standard exponential bound for a sum of independent bounded variables [6] gives $\mathbb{P}[|\Lambda| > q/3] \le e^{-q/72}$. This bound follows, for example, from $\mathbb{E}e^{t(B-\mathbb{E}B)} \le e^{t^2/8}$ for a Bernoulli variable B: multiply over the q coordinates and optimize at $t = 1/3$. Character expansion on $\mathbb{F}_2 \times \mathbb{F}_2^n$ gives exactly, with $b = (1-2p)^{q/2}$,

$$\mathbb{P}[(\epsilon, \ell) = (0,0)] = \frac{1 + (1-2p)^q + (2q-2)(1-2p)^{q/2}}{2q} = b + \frac{(1-b)^2}{2q} \le (1-p)^q + \frac{1}{2q}, \tag{21}$$

where the inequality uses $1-2p \le (1-p)^2$, hence $b \le (1-p)^q$, and $0 \le b \le 1$. The parity-only character uses all q coordinates; every character with a nonzero linear part uses $q/2$ coordinates. These are all $2q$ characters, proving the formula and (18). When $p \to 0$ and $qp \to \lambda < \infty$, $(1-p)^q \to e^{-\lambda}$ and the additive error in (18) vanishes, proving (19). If $qp \to \infty$, then $(1-p)^q \le e^{-qp} \to 0$. □

The error in (18) is additive; when $qp \to \infty$ it gives no relative control, and we do not claim $\mathbb{P}[\mathcal{C}_T(U)] \sim (1-p)^q$ in that regime. Since $qp = 2^{n-\rho+1}$, a finite positive limit λ in (19) necessarily lies in $2^{\mathbb{Z}}$ in the present fair-bit model.

Permuting coordinates preserves the role distribution and the target line T, so in applying Theorem 5.2 the reference coordinate may be chosen freely. For this family, taking a coordinate equal to 1 as reference leaves exactly one ratio different from 1, so $s = 1$. In any case this family has exactly two distinct slopes, so $\nu = 1$ and $\mathcal{A} = qp$ for every choice of reference coordinate. Theorem 5.2 thus already gives vanishing orbit constancy when $qp \to \infty$, and the lower bound in (18) shows that this condition is necessary. Hence the scale $qp = 2^{n-\rho+1}$ in Theorem 5.2 cannot be improved for general coordinate-full directions, and Theorem 6.1 identifies the critical asymptotics for this family.

7. Positive association and high-density orbit bounds

Return to the four-point coefficient K of (3). The unit A in this section is supported on a single line M, whereas the target line T may be any line in G. No coordinate-full assumption is needed for the following lower bounds. Put $\theta = 2^{-\rho}$.

Lemma 7.1 (Product-measure association). If $E_1, \dots, E_m$ are decreasing events of finitely many independent bits, then $\mathbb{P}[\bigcap_i E_i] \ge \prod_i \mathbb{P}[E_i]$.

Proof. For two decreasing functions f, g, induct on the number of bits in the identity

$$\operatorname{Cov}(f,g) = \mathbb{E}\operatorname{Cov}(f,g \mid Z) + \operatorname{Cov}(\mathbb{E}[f \mid Z], \mathbb{E}[g \mid Z]),$$

where Z is the last bit. The first term is nonnegative by induction. If $\mathbb{P}[Z=1] = r$, the second is $r(1-r)(f_1-f_0)(g_1-g_0) \ge 0$, where $f_z = \mathbb{E}[f \mid Z = z]$ and similarly for g. This proves the two-event case. The intersection of decreasing events is decreasing, so induction on m completes the proof. This is the product-measure form of positive association [2], [3]. □

Theorem 7.2 (Four-point orbit lower bounds). For every fixed $A \in \mathbb{F}_2[M]$, with $s_A = |\operatorname{supp}(A)|$,

$$\mathbb{P}_X[\mathcal{C}_T(KA)] \ge (1-\theta)^{4qs_A} \ge (1-\theta)^{4q^2}. \tag{22}$$

More precisely, with $\eta = |H \cap (M+T)|$,

$$\mathbb{P}_X[\mathcal{C}_T(KA)] \ge (1-\theta)^{\min\{4qs_A,\, 4|M+T|/\eta\}}. \tag{23}$$

In particular, if $M, T \le W$ are distinct, then $\mathbb{P}_X[\mathcal{C}_T(KA)] \ge (1-\theta)^{q^2}$ for every $A \in \mathbb{F}_2[M]$. For A uniform on $\mathcal{U}(M)$, put $r = (1-\theta)^{4q}$. Then

$$\mathbb{E}_A \mathbb{P}_X[\mathcal{C}_T(KA)] \ge L_q(r) := 2^{-q}\big((1+r)^q - (1-r)^q\big). \tag{24}$$

If $M = T$, then for every $A \in \mathcal{U}(T)$ the events $\mathcal{C}_T(KA)$ and $\mathcal{C}_T(K)$ are equal, and with $h = |H \cap T|$,

$$\mathbb{P}_X[\mathcal{C}_T(KA)] \ge (1-\theta)^{4q/h}. \tag{25}$$

Proof. The union of the supports of all $\tau_t(KA)$, $t \in T$, contains at most $4qs_A$ points, and it is contained in the subgroup $H + M + T$, whose order is $|H||M+T|/|H \cap (M+T)| = 4|M+T|/\eta$. For each such point g, the event $\{\prod_j X_{j,g_j} = 0\}$ is decreasing and has probability $1-\theta$. If all these events occur, the entire orbit is zero. Lemma 7.1 proves (22) and (23). If $M, T \le W$ are distinct lines, then $M + T = W \supseteq H$, so $\eta = 4$ and the exponent in (23) is at most q^2. For a uniform augmentation-one coefficient on a q-point line, the support size has the binomial distribution conditioned to be odd. Therefore

$$\mathbb{E}_A r^{s_A} = 2^{-(q-1)} \sum_{k \text{ odd}} \binom{q}{k} r^k = 2^{-q}\big((1+r)^q - (1-r)^q\big),$$

which proves (24).

For the same-line statement, let $I(T)$ be the augmentation ideal of $\mathbb{F}_2[T]$. The orbit event $\mathcal{C}_T(B)$ is equivalent to $Q_{eB}(X) = 0$ for every $e \in I(T)$, since the elements $1 + [t]$ span that ideal. Multiplication by a unit $A \in \mathcal{U}(T)$ maps $I(T)$ bijectively

onto itself. Thus $I(T)KA = I(T)K$, giving equality of the events pointwise in X. Apply the first part of the proof directly to K: the support union is $H + T$, of size $4q/h$. □

For the individual coefficient A, the first bound in (22) strengthens the union bound $1 - 4qs_A\theta$. Replacing s_A by q is a uniformization and need not strengthen that individual union bound.

Corollary 7.3. If $\rho = 2n + c_n$ and $c_n \to c$, then the two lower bounds in (22) after uniformization and (24) tend respectively to

$$e^{-4\cdot 2^{-c}} \quad \text{and} \quad e^{-2\cdot 2^{-c}}. \tag{26}$$

If $\rho - 2n \to +\infty$, then $\mathbb{P}_X[\mathcal{C}_T(KA)] \to 1$ uniformly over all line-supported A. If $M, T \le W$ are distinct, the bound $(1-\theta)^{q^2}$ from (23) holds for every $A \in \mathbb{F}_2[M]$ without averaging and tends to $e^{-2^{-c}}$, which exceeds both constants in (26). For a same-line unit, the lower bound in (25), at $\rho = n + c_n$ with $c_n \to c$ and fixed h, tends to $e^{-4\cdot 2^{-c}/h}$.

Proof. At $\rho = 2n + c_n$, $q^2\theta \to 2^{-c}$, so $(1-\theta)^{4q^2} \to e^{-4\cdot 2^{-c}}$. Also $1 - r \sim 4q\theta$ and $((1+r)/2)^q \to e^{-2\cdot 2^{-c}}$, whereas $((1-r)/2)^q \to 0$. The remaining assertions follow from the same logarithmic expansion of $\log(1-\theta)$. □

These are lower bounds for the all-zero subevent. The scale $\rho = 2n$ is a threshold of the displayed support lower bound; it is not a determination of the true cross-line orbit transition. The equalities for survival in Section 4 do not change that distinction.

8. Exact verification over $\mathbb{F}_{16}$

We checked the exposed-column count of Lemma 5.1 with $q = 16$, $\rho = 3$ and $E = \{0, 1\}$. Represent the field as $\mathbb{F}_2[z]/(z^4 + z + 1)$, label its elements by polynomial bits, and normalize the last slope coordinate to one. The ordered parameter set is

$$\{(d_1, d_2, 1) : d_1, d_2 \in \mathbb{F}_{16}^*,\ (d_1, d_2) \neq (1, 1)\},$$

of cardinality $15^2 - 1 = 224$. For each slope we tested the three nonzero relations on E. The exposed set was fixed by increasing-label greedy selection in the graph (13), with all pairs of distinct rows in E used to construct the graph before the relation was examined. This specifies all 672 cases without symmetry reductions.

To avoid enumerating 2^{32} input pairs per case, condition on the first role array $A \in \mathbb{F}_2^{16}$. For the second array B, the exposed-column equations form a homogeneous binary linear system $D_A B = 0$, where

$$D_A(x, u) = \sum_{a \in E} c_a \sum_{\substack{g \neq 0 \\ a+g=x}} A_{a+d_1 g} \mathbf{1}_{u = a + d_2 g}, \qquad x \in F. \tag{27}$$

The number of second arrays is exactly $2^{16 - \operatorname{rank}_{\mathbb{F}_2} D_A}$, so the number of vanishing input pairs is

$$N = \sum_{A \in \mathbb{F}_2^{16}} 2^{16 - \operatorname{rank}_{\mathbb{F}_2} D_A}. \tag{28}$$

The denominator of the event probability is 2^{32}.

The computation constructs (27) directly from nonzero line parameters g. It separately checks the affine indices in (9), including its zero diagonal. An independent reference enumerates the at most four role-labelled bits in each exposed column. The program checks that distinct columns use disjoint sets of variables before multiplying their exact local probabilities. Neither this reference nor the rank count is derived from the proposed upper bound. A full input-pair check over $\mathbb{F}_4$ additionally verifies the conditional-rank count.

All 672 exact counts agree with the reference and satisfy $N 4^{|F|} \le 2^{32} 3^{|F|}$. Table 1 groups the complete output; m denotes the relation support size. The largest ratio of the measured probability to $(3/4)^{|F|}$ is exactly one. All cases are resolved in this specified finite domain.

Table 1. Complete exact exposed-column counts for $q = 16$, $\rho = 3$, $E = \{0, 1\}$. Every probability is $N/2^{32}$.

s	m	$\lvert F\rvert$	Cases	N	Probability
1	1 or 2	8	84	429981696	6561/65536
2	1	6	112	764411904	729/4096
2	1	8	280	429981696	6561/65536
2	2	6	56	256000000	15625/262144
2	2	8	140	100000000	390625/16777216

This calculation checks the field indexing and the exact count. The general theorem is proved by Lemma 5.1 and Theorem 5.2; the finite calculation is not an asymptotic argument. The supplementary material contains the C++17 program, the exact per-case rank histograms and reproduction commands.

9. Limitations and open questions

For a singleton and one fixed transverse line-sum unit, Theorem 5.2 establishes vanishing orbit constancy at every fixed density $\kappa = \lim \rho/n < 1$. The repeated-slope family in Theorem 6.1 has the exact critical law $e^{-\lambda}$ for $\lambda = \lim 2^{n-\rho+1}$. The generic $\kappa = 1$ window remains an open problem. A formulation should specify the slope sequence or the sampling law for slopes and determine which parameters of the slope multiplicities control the limiting orbit probability. A repeated-slope limit alone does not classify directions lacking that structure.

For a four-point coefficient multiplied by a uniform line unit, the cross-line orbit probability has a positive lower bound at $\rho = 2n + O(1)$ and tends to one above that scale. Determining its actual transition below this sufficient saturation scale requires more information than the support union. Same-line units preserve the orbit event exactly and already have a positive lower bound at $\rho = n + O(1)$. An exact survival probability can therefore coexist with substantially different orbit behavior as the number of roles changes.

Theorem 4.3 determines all products restricted to the three defining lines, including arbitrary repetitions. For three general supporting lines in W, the product equation among the quotient subalgebras is still a well-defined finite counting problem, but the Cartesian-support proof need not apply. An exact law for that larger family is a separate open question.

Reproducibility

All proofs are self-contained. The supplementary archive contains (a) a standard-library Python script that checks by exhaustive enumeration, over $\mathbb{F}_4$, $\mathbb{F}_8$ and (for the rank statements of Section 6) $\mathbb{F}_{16}$, the fixed-coefficient identity and the minimum-support claim, the one-, two- and three-line survival laws with repetitions, the exact exposed-column probability, the rank and orbit upper bounds, the identities used in the repeated-slope proof (including the exact parity/sum formula), and the association lower bounds and the pointwise same-line equality of events, in small parameter ranges; and (b) the C++17 program and complete output for the $q = 16$ count of Section 8. These are finite numerical checks, not proofs.

References

1. M. Borello, W. Willems and G. Zini, *On ideals in group algebras: An uncertainty principle and the Schur product*, Forum Mathematicum (2022). DOI: 10.1515/forum-2022-0064; preprint arXiv:2202.12621.
2. T. E. Harris, *A lower bound for the critical probability in a certain percolation process*, Proceedings of the Cambridge Philosophical Society 56 (1960), 13–20. DOI: 10.1017/S0305004100034241.
3. C. M. Fortuin, P. W. Kasteleyn and J. Ginibre, *Correlation inequalities on some partially ordered sets*, Communications in Mathematical Physics 22 (1971), 89–103. DOI: 10.1007/BF01651330.
4. R. W. R. Darling, M. D. Penrose, A. R. Wade and S. L. Zabell, *Rank deficiency in sparse random GF[2] matrices*, Electronic Journal of Probability 19 (2014), 1–36. DOI: 10.1214/EJP.v19-2458.
5. I. S. Reed and G. Solomon, *Polynomial codes over certain finite fields*, Journal of the Society for Industrial and Applied Mathematics 8 (1960), 300–304. DOI: 10.1137/0108018.
6. W. Hoeffding, *Probability inequalities for sums of bounded random variables*, Journal of the American Statistical Association 58 (1963), 13–30. DOI: 10.1080/01621459.1963.10500830.